V9: Cross-Model, Cross-Domain Neural Basis of Risky Decision-Making in LLMs

Internal Research Report (v9.5)

"Can Large Language Models Develop Gambling Addiction?"

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Models: Gemma-2-9B-IT (42L, 3584-dim, GemmaScope 131K) | LLaMA-3.1-8B-Instruct (32L, 4096-dim, LlamaScope 32K)
Paradigms: Investment Choice (IC, 1600 games), Slot Machine (SM, 3200 games), Mystery Wheel (MW, 3200 games)
Data: Gemma: IC+SM+MW; LLaMA: IC+SM (MW pending)
Pipeline: StandardScaler → PCA(50) → LogReg(C=1.0, balanced) | 5-fold StratifiedKFold

Data integrity: All numerical claims are traceable to computed JSON outputs.

Central Question

Do different models, across different gambling tasks, under different prompt conditions, share a common neural basis for bankruptcy — the most extreme risk outcome?

- **RQ1**: Do Gemma and LLaMA share common activation/feature patterns that predict bankruptcy?
- **RQ2**: Do activation patterns remain invariant when the domain (IC, SM, MW) changes?
- **RQ3**: Are neural representations consistent across prompt conditions (Fixed/Variable, G/M/W/P/R)?

Executive Summary

RQ1 — Universal BK Features Exist Across Architectures. 600 of 3,584 Gemma L22 neurons (16.7%) are sign-consistent universal BK predictors across three paradigms (FDR $p < 0.01$). 744 SAE features across 7 layers show cross-domain consistency (L18+ binomial $p < 5 \times 10^{-3}$). LLaMA achieves near-identical BK classification (AUC 0.974 vs Gemma 0.976 in SM). Factor decomposition confirms 65–71% of SAE features encode BK independently of bet-type and paradigm (permutation $p = 0.000$, null ∼1%).

RQ2 — Strong Domain Invariance (Gemma), Selective Invariance (LLaMA). Gemma cross-domain transfer reaches AUC 0.87–0.93 (IC→MW, SM→MW; all $p = 0.000$). A 3D shared BK subspace achieves AUC 0.86–0.97 despite near-orthogonal LR weight vectors. LLaMA IC→SM transfer succeeds only at L25 (AUC 0.783), while SM→IC is more stable (AUC 0.685 at L30). Hidden state transfer exceeds SAE transfer by 0.10–0.25 in most directions.

RQ3 — BK Representation Is Condition-Invariant, but Condition Effects Are Model-Dependent. LLaMA IC shows cos > 0.81 between Variable and Fixed BK directions across all layers ($n \geq 65$ per group). Both models have ∼33–37% shared BK neurons. G(Goal) dominates BK behaviorally in Gemma (20.75×) but not LLaMA (1.26×), yet both share the mechanism: G drives BK independently without modulating existing BK representations.

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1 Setup

1.1 Data & Representations

Table 1: Dataset Overview

Paradigm	Games	BK	BK%	Bet Types	Constraints	Prompt Cond.
IC (Gemma)	1,600	172	10.8%	Fixed/Var	c10/c30/c50/c70	BASE/G/M/GM
SM (Gemma)	3,200	87	2.7%	Fixed/Var	$10 fixed	32 combos
MW (Gemma)	3,200	54	1.7%	Fixed/Var	$30 fixed	32 combos
IC (LLaMA)	1,600	142	8.9%	Fixed/Var	c10/c30/c50/c70	BASE/G/M/GM
SM (LLaMA)	3,200	1,164	36.4%	Fixed/Var	$10 fixed	32 combos

1.2 V8 Limitations Addressed

V8 (Gemma-only) left three weaknesses:

1. **SAE cross-domain failure**: Only 1/131K features was sign-consistent at L22 top-50. V9: **744 features** across 7 layers (§2.2).
2. **Variable/Fixed "dissociation"**: Artifact of comparing all games, not BK-only (§4.1).
3. **Single model**: V9 adds LLaMA IC+SM (§2.3, §3.5).

1.3 Cross-Model BK Rate Asymmetry

Table 2: BK Rate Asymmetry Between Models

	Gemma IC	LLaMA IC	Gemma SM	LLaMA SM
Fixed BK rate	**19.75%**	8.1%	—	—
Variable BK rate	1.75%	**9.6%**	—	—
Overall BK rate	10.8%	8.9%	2.7%	**36.4%**
Autonomy effect	−18.0pp	+1.5pp	—	—

2 RQ1: Universal BK Features

2.1 600 Universal BK Neurons (Gemma L22)

Point-biserial correlation per neuron per paradigm, FDR-corrected (BH, $p < 0.01$). Neurons significant in all three paradigms with consistent sign are "Universal BK Neurons."

Table 3: Universal BK Neuron Summary (Gemma L22)

	IC	SM	MW	Cross-paradigm
FDR-significant	2,528 (70.5%)	2,725 (76.0%)	2,314 (64.6%)	—
All-3-significant	—	—	—	1,238 (34.5%)
Sign-consistent	—	—	—	**600 (16.7%)**
BK-promoting	—	—	—	302
BK-inhibiting	—	—	—	298

Table 4: Top-5 Universal BK Neurons

Rank	Neuron	min$\|r\|$	IC r	SM r	MW r	Direction
1	**1763**	0.217	+0.305	+0.248	+0.217	BK-promoting
2	371	0.203	−0.223	−0.203	−0.209	BK-inhibiting
3	2951	0.198	−0.198	−0.207	−0.266	BK-inhibiting
4	1755	0.190	−0.210	−0.198	−0.190	BK-inhibiting
5	864	0.182	−0.267	−0.182	−0.182	BK-inhibiting

Neuron #1763 is the strongest universal BK-promoting neuron (min$|r| = 0.217$). Promoting (302) $\approx$ Inhibiting (298) at L22 — balanced representation. This exactly reproduces V8.

2.2 744 SAE Features (Gemma, 7 Layers)

V9 tests 7 layers (L10, 12, 18, 22, 26, 30, 33) using all active features. Sign-consistent: same Cohen's d sign across IC/SM/MW. Strong: geometric mean $|d| \geq 0.2$.

Table 5: Gemma Multi-Layer SAE Cross-Domain Consistency

Layer	Active	Sign-consistent	Strong ($d \geq 0.2$)	% consistent
L10	92	30	23	32.6%
L12	112	35	31	31.2%
L18	170	58	52	34.1%
L22	281	109	95	38.8%
L26	303	133	119	43.9%
L30	415	188	173	45.3%
L33	456	191	172	41.9%
Total	—	**744**	**665**	—

Binomial test (3-paradigm chance = 25%): L10–L12 are NS. L18+ significantly exceed chance ($p < 5 \times 10^{-3}$).

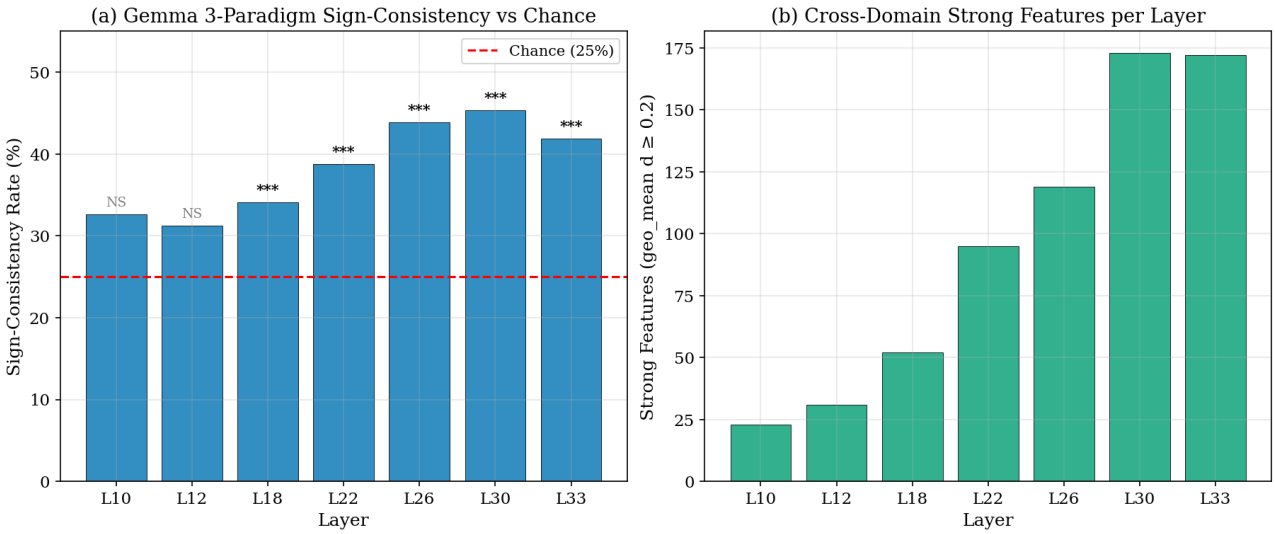

Figure 1: Gemma 3-paradigm sign-consistency with binomial test. L10–L12 do not exceed chance (25%, dashed). L18+ far exceeds chance. Strong features peak at L30 (173).

2.3 Cross-Model Comparison

Table 6: BK Classification AUC (DP, SAE → PCA → LogReg)

Paradigm	Gemma-2-9B	LLaMA-3.1-8B
SM best AUC	**0.976** (L20)	**0.974** (L8)
IC best AUC	**0.960** (L30)	**0.954** (L12)

Table 7: LLaMA SM AUC Across Layers

Layer	AUC	n_features	n_BK
L0	0.970	157	1,164
L4	0.971	750	1,164
L8	**0.974**	818	1,164
L12	0.973	854	1,164
L15	0.970	962	1,164
L20	0.963	1,028	1,164
L25	0.961	1,182	1,164
L30	0.959	1,368	1,164

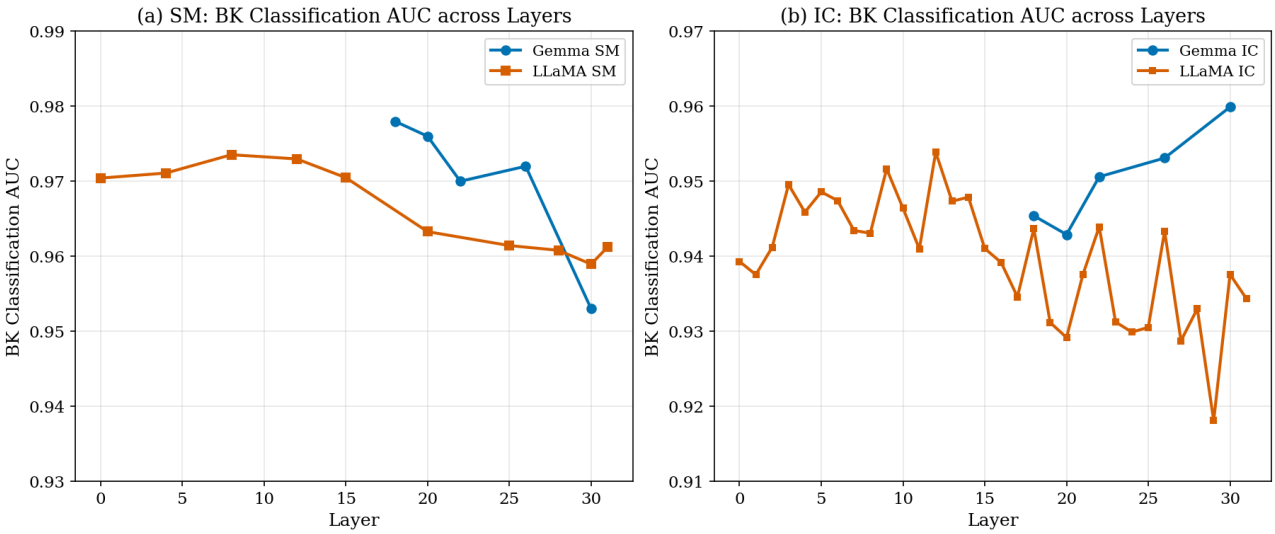

Figure 2: BK classification AUC across layers. (a) SM: both models AUC > 0.95. (b) IC: Gemma peaks L30 (0.960), LLaMA peaks L12 (0.954). Encoding depth differs but performance is near-identical.

2.4 RQ1 Synthesis

1. **Near-identical BK prediction**: 0.976 vs 0.974 (SM) — architecture-invariant information content.
2. **BK-inhibiting dominance at L30**: Both models (all $p < 0.001$). Not at L22 (LLaMA SM: $p = 0.897$).
3. **Encoding depth differs**: Gemma peaks mid-deep (L20), LLaMA peaks early (L8).
4. **R1 early prediction**: AUC 0.665–0.801 from Round 1.
5. **Factor decomposition**: 65–71% features encode BK independently (perm $p = 0.000$).

3 RQ2: Domain Invariance

3.1 Hidden State Cross-Domain Transfer (Gemma L22)

Table 8: Hidden State vs SAE Transfer (Gemma L22)

Transfer	Hidden AUC	SAE AUC	Δ (HS−SAE)	HS perm_p
IC → SM	**0.746**	0.499 (NS)	+0.247	0.000
IC → MW	**0.826**	0.876	−0.050	0.000
SM → MW	**0.920**	0.819	+0.101	0.000

Hidden state transfer exceeds SAE in IC→SM (+0.247) and SM→MW (+0.101). BK signal is distributed; SAE sparsification loses coherence.

3.2 SAE Cross-Domain Transfer (Gemma)

Table 9: Best SAE Transfer AUC Per Direction (Gemma)

Transfer	Best Layer	AUC	perm_p	Null mean
IC → SM	L26	**0.913**	0.000	0.498
IC → MW	L18	**0.932**	0.000	0.504
SM → MW	L30	**0.867**	0.000	0.504
SM → IC	L30	0.646	0.000	0.500
MW → IC	L22	0.853	0.000	0.500

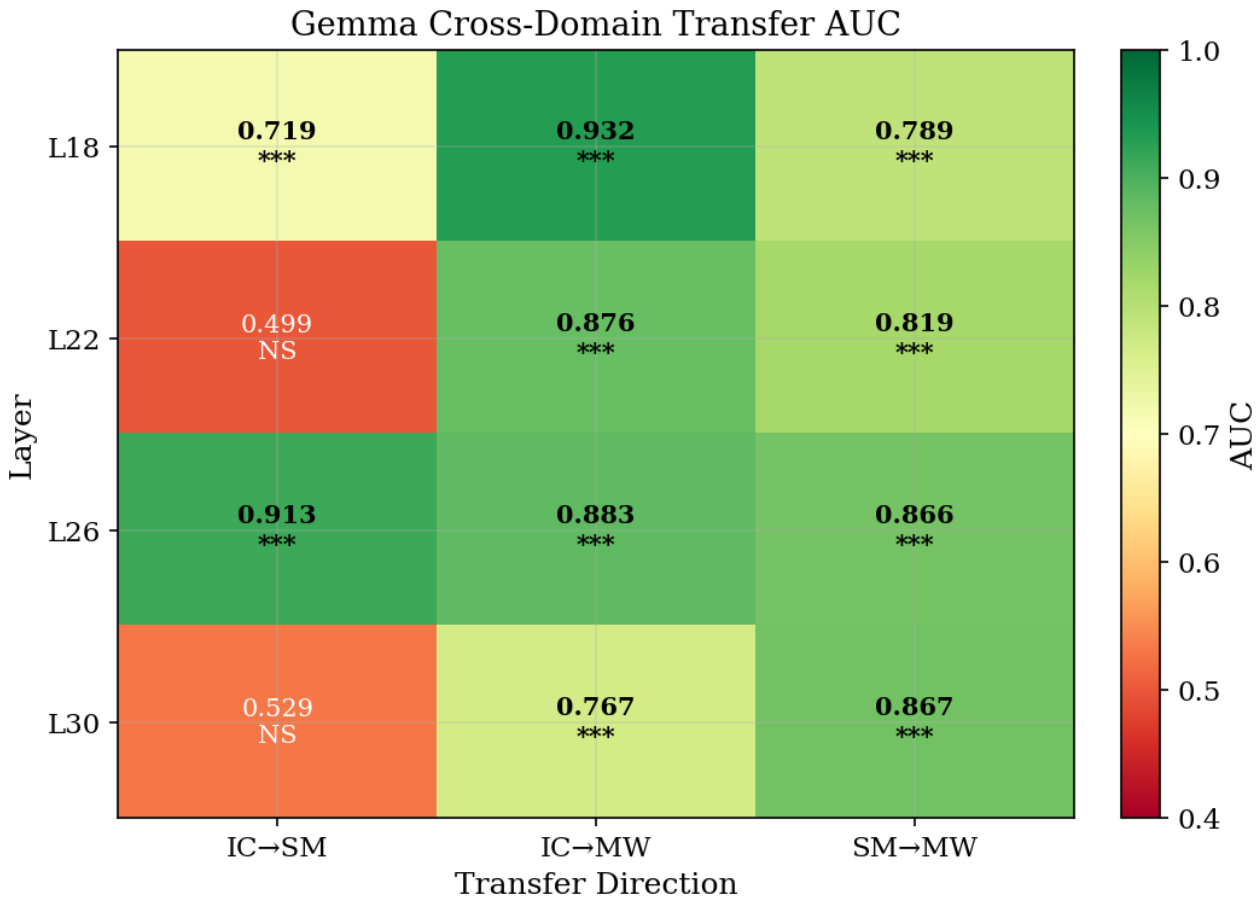

Figure 3: Gemma cross-domain transfer heatmap. IC→MW and SM→MW are significant at all layers. IC→SM is layer-dependent (L26: 0.913, L22: 0.499 NS).

3.3 3D Shared BK Subspace (Gemma L22)

Table 10: 3D Shared Subspace Performance

Paradigm	3D Subspace AUC	Full-dim AUC (50-PC)
IC	0.862 ± 0.029	∼0.95+
SM	0.899 ± 0.034	∼0.97+
MW	0.970 ± 0.016	∼0.97+

LR weight vector cosines: IC–SM=0.042, IC–MW=−0.026, SM–MW=−0.026 (near-orthogonal). Despite this, the 3D subspace achieves AUC 0.86–0.97 — BK signal is distributed across a subspace, not a single direction.

3.4 Factor Decomposition

Table 11: Factor Decomposition: Outcome-Significant Features

Factor	Gemma (581 feat., 3-par.)	LLaMA (1,418 feat., 2-par.)
Outcome (BK)	**65.2%**	**69.5%**
Bet-type	92.3%	81.4%
Paradigm	99.8%	96.2%

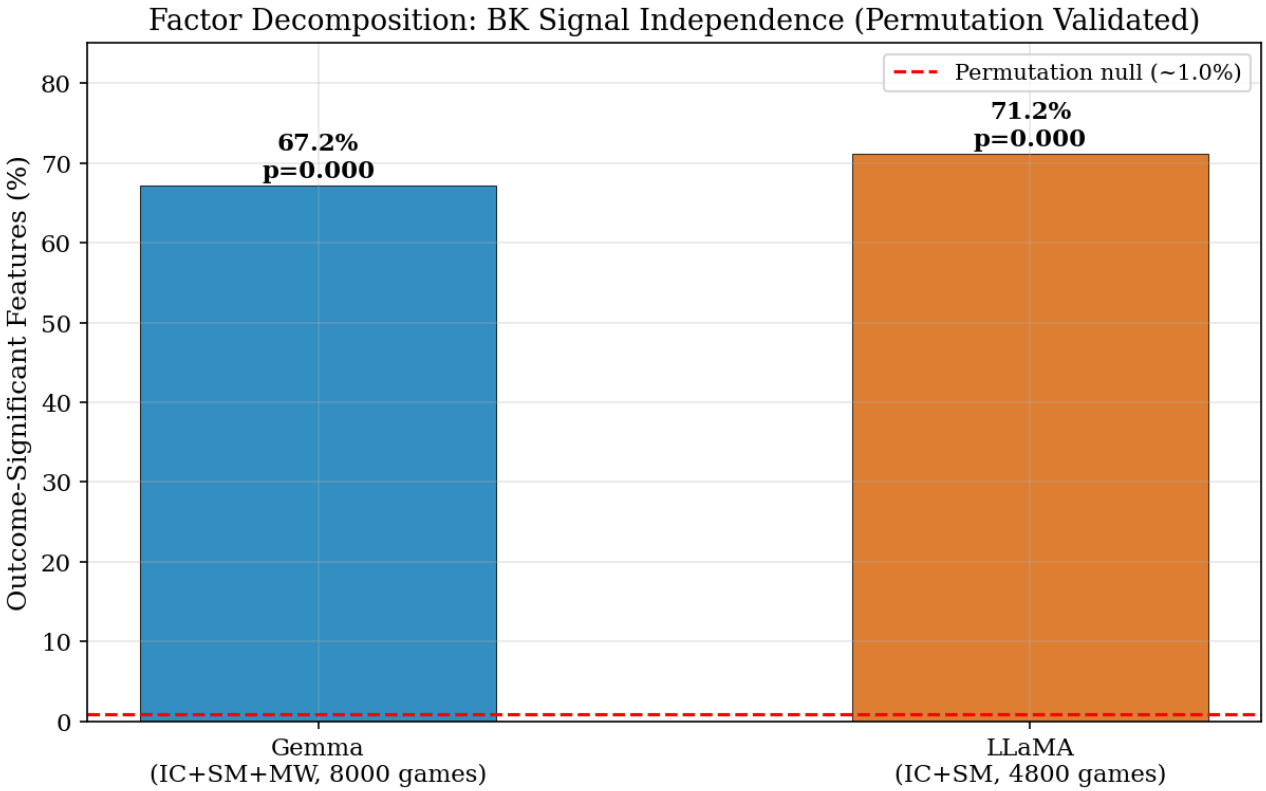

Figure 4: Factor decomposition with permutation test. Both models: 65–71% outcome-significant (blue/orange), exceeding null (∼1%, dashed) by 60×+. Perm $p = 0.000$.

3.5 LLaMA Cross-Domain Transfer (IC ↔ SM)

Table 12: LLaMA IC↔SM Transfer

Transfer	Layer	AUC	perm_p	Null mean
IC → SM	L25	**0.783**	0.000	0.500
SM → IC	L30	**0.685**	0.000	0.498
SM → IC	L12	0.662	0.000	0.498
SM → IC	L8	0.624	0.000	0.499

Asymmetric: SM→IC works at multiple layers; IC→SM succeeds only at L25. Mirrors Gemma (IC→SM layer-dependent at L26).

3.6 LLaMA Sign-Consistency & Within-Bet-Type

LLaMA 2-paradigm sign-consistency: 51–57% per layer, but chance level is 50%. Only 3/16 layers significantly exceed chance. Raw sign-consistency is weak evidence; transfer AUC (§3.5) is more reliable.

LLaMA within-bet-type: Fixed d-correlation $= 0.209$ ($p = 1.2 \times 10^{-7}$) confirms Fixed cross-domain consistency. Variable d-correlation $= -0.097$ — IC and SM Variable BK representations are genuinely different (bootstrap: mean$=-0.075$, 95% CI $[-0.179, +0.032]$).

4 RQ3: Prompt Condition Effects

4.1 Fixed vs Variable BK Direction

Table 13: BK Direction Cosine Similarity (Gemma IC, 3584 neurons)

Layer	Var BK	Fix BK	$\cos(\text{dir}_{\text{Var}}, \text{dir}_{\text{Fix}})$	Common neurons
L10	14	158	−0.195	178
L18	14	158	−0.082	197
L22	14	158	+0.330	555
L26	14	158	+0.443	1,053
L30	14	158	+0.401	1,073

Table 14: BK Direction Cosine Similarity (LLaMA IC, 4096 neurons)

Layer	$\cos(\text{dir}_{\text{Var}}, \text{dir}_{\text{Fix}})$	Common neurons
L8	**0.882**	2,513 (61.4%)
L12	**0.814**	2,424 (59.2%)
L22	**0.835**	2,587 (63.2%)
L25	**0.837**	2,545 (62.1%)
L30	**0.819**	2,488 (60.7%)

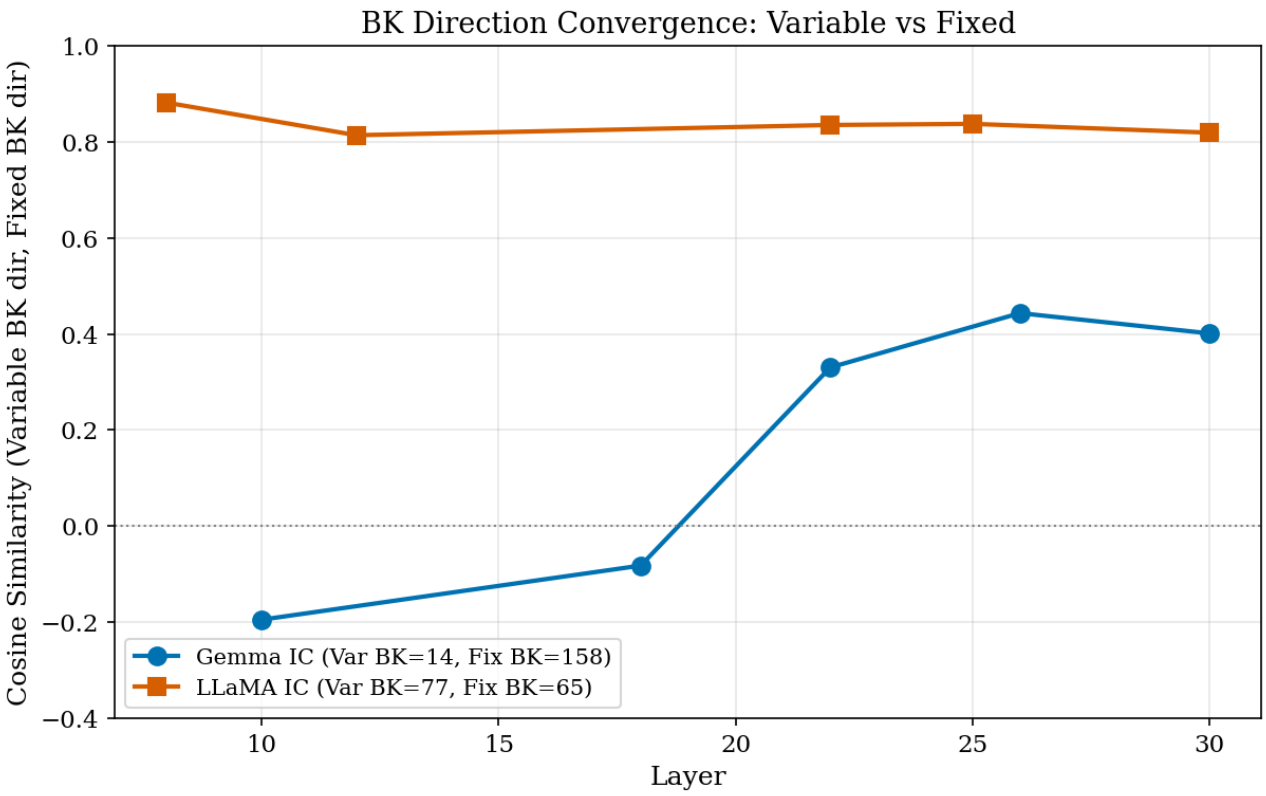

Figure 5: BK direction cosine across layers. Gemma (blue): negative at shallow, positive at deep. LLaMA (orange): cos > 0.81 at all layers. Gemma's shallow negative is likely a sample-size artifact (Var BK=14).

4.2 Shared BK Neurons (Interaction Regression)

Table 15: Shared BK Neurons Across Models

	Gemma IC L22	LLaMA IC L22	LLaMA SM L22
n_neurons	3,584	4,096	4,096
n_BK	172	142	1,164
Outcome-sig	1,639 (45.7%)	3,150 (76.9%)	1,052 (25.7%)
Interaction-sig	448 (12.5%)	1,605 (39.2%)	1,495 (36.5%)
Shared BK	**1,182 (33.0%)**	**1,501 (36.6%)**	**72 (1.8%)**

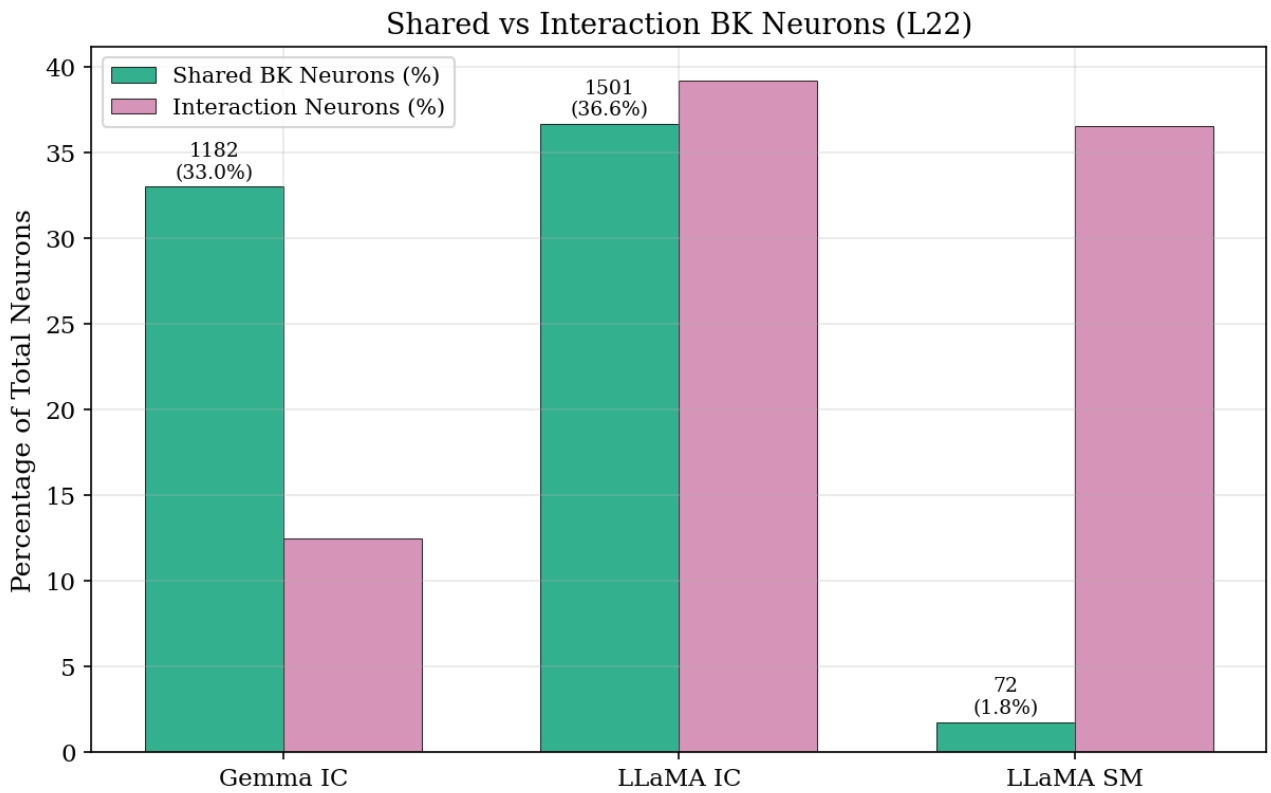

Figure 6: Shared vs Interaction BK neurons. Gemma IC (33.0%) and LLaMA IC (36.6%) show similar proportions. LLaMA SM (1.8%) reflects Variable-dominated BK structure.

4.3 Prompt Component Analysis

Table 16: Prompt Component BK Rate Effects (Cross-Model)

Component	Gemma SM ratio	LLaMA SM ratio	Cross-model
G (Goal)	**20.75**×	1.26×	Gemma $\gg$ LLaMA
M (Mood/Money)	2.35×	1.31×	Gemma > LLaMA
W (Warning)	2.35×	1.15×	Gemma > LLaMA
R (Reset)	1.72×	—	Gemma only
P (Persona)	1.17×	1.06×	Both negligible

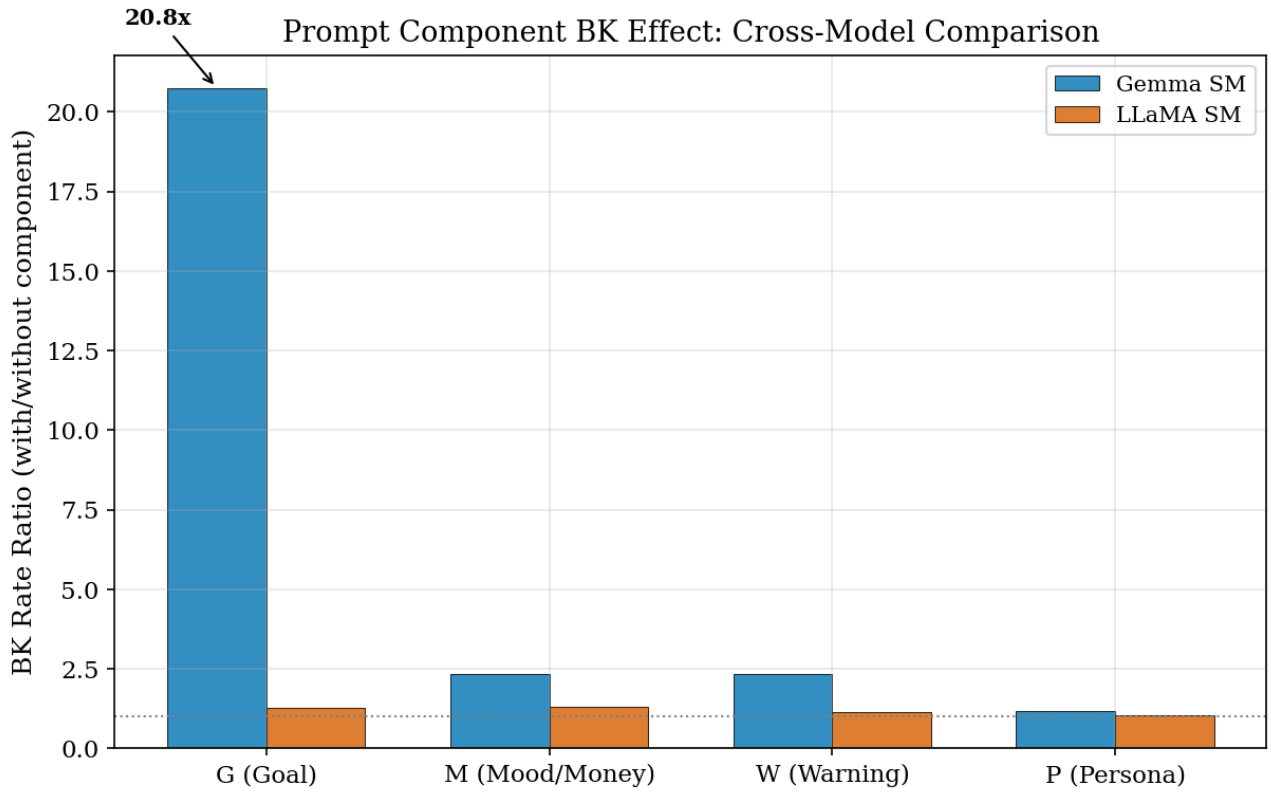

Figure 7: Prompt component BK effect. G(Goal) produces 20.8× BK increase in Gemma (dominant) but only 1.26× in LLaMA. Same prompt, 16× different behavioral effect.

G drives BK independently in both models (amplifies_bk ≈ 0 in Gemma, 8.2% in LLaMA). Warning effects are shallow-only in both models (vanishes at deep layers). M(Money) is LLaMA's highest BK amplifier (14.5% at L30), negligible in Gemma.

4.4 G-Prompt Alignment & Bet Constraint

G-Prompt direction alignment (Gemma L22): SM $\cos(G, BK) = +0.850$ (strong), IC $= -0.143$, MW $= +0.367$. G pushes hidden states toward BK in SM.

Bet constraint linear mapping (Gemma IC L22): c10→c70 maps linearly to BK-projection ($r = 0.979, p = 0.021$). c10: 0% BK, c70: 21% BK.

5 Limitations & Critical Assessment

5.1 Validated vs. Unvalidated Claims

Table 17: Permutation-Validated Claims

Claim	Test	Result
Factor decomposition 67–71%	100-perm shuffle	$p = 0.000$ ✓
BK-inhibiting L30	Binomial	All $p < 0.001$ ✓
BK-inhibiting L22	Binomial	LLaMA SM $p = 0.897$ ×
Variable cross-domain	Bootstrap	mean=−0.075, CI incl. 0 ×
Classification AUC	100-perm shuffle	$p = 0.000$ ✓
Gemma 3-par. consistency	Binomial vs 25%	L18+ $p < 5 \times 10^{-3}$ ✓

5.2 Key Limitations

1. **LLaMA MW not extracted**: 3-paradigm cross-model comparison not possible.

2. **Gemma IC Variable BK=14**: Very limited power for bet-type comparisons.
3. **No causal validation**: All analyses are correlational.
4. **Multiple comparisons**: FDR not systematically applied to all analyses.
5. **Cross-model effect size confound**: Gemma SM $n_{BK} = 87$ vs LLaMA $n_{BK} = 1{,}164$ — small-sample inflation in Gemma.
6. **BK rate heterogeneity**: 2.7% vs 36.4% across models complicates comparison.

6 Next Steps

1. **Activation patching**: Causal validation of top cross-domain features.
2. **LLaMA MW extraction**: Complete 3-paradigm comparison.
3. **IC→SM transfer mechanism**: Investigate layer-specific alignment.
4. **FDR correction**: Apply to all per-feature/per-neuron tests.
5. **G component deep dive**: Explain 20.75× vs 1.26× divergence.
6. **BK rate confound control**: Balanced subsampling for IC-SM re-analysis.